



The Real Architecture Behind Vionto's AI Pipeline

How a photo album becomes a directed, narrated, deterministic video - and where generative AI is deliberately kept optional.



THE CORE IDEA

AI interprets and directs. FFmpeg renders the final frames. Optional generated motion clips are reviewed and accepted before they can replace a still-image segment.

EXECUTIVE VIEW

An AI director wrapped around a deterministic renderer

Vionto is a photo-to-video system inside the ASafariM monorepo. The browser owns the creative workflow; server routes assemble trusted data; AI services interpret media and produce narrative assets; a separate BullMQ worker turns a validated manifest into an MP4 with FFmpeg.

ONE SENTENCE

AI chooses meaning and language; deterministic software renders the frames.

01

AUTHORING

Next.js UI, project / album / version settings, story edits, clip approval

02

INTELLIGENCE

Vision captions, EXIF context, scoring, clustering, story generation, TTS

03

PRODUCTION

Server-built Zod manifest, Redis queue, worker, ffprobe, FFmpeg, export

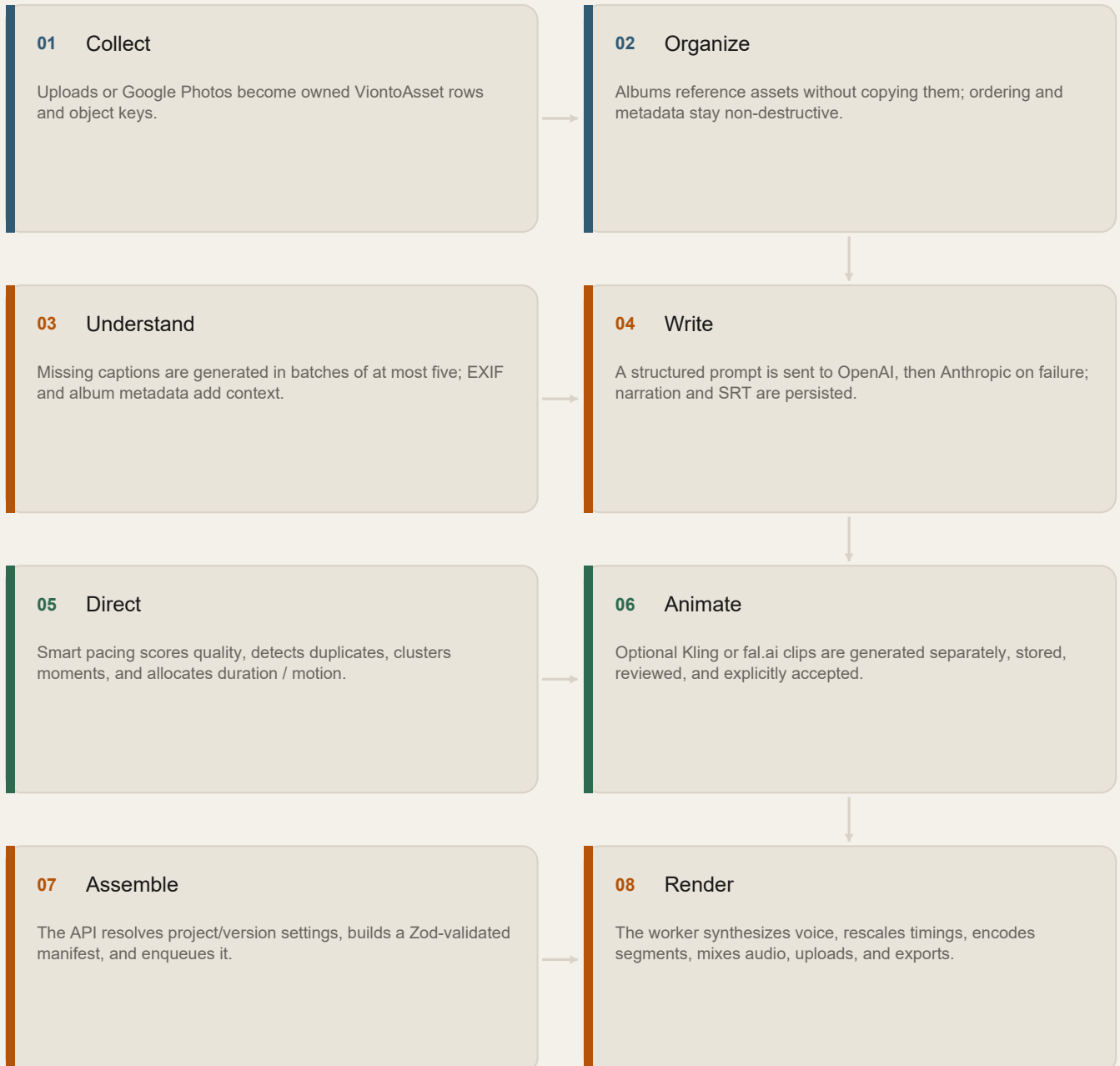
Architectural promises

Provider-neutral: vendor SDKs stay behind adapters. Trusted boundary: the server builds the manifest. Durable media: generated clips are copied out of temporary provider URLs. Bounded work: 1-200 visual assets and up to 16 audio tracks per manifest.

END-TO-END

From album to export: the real request path

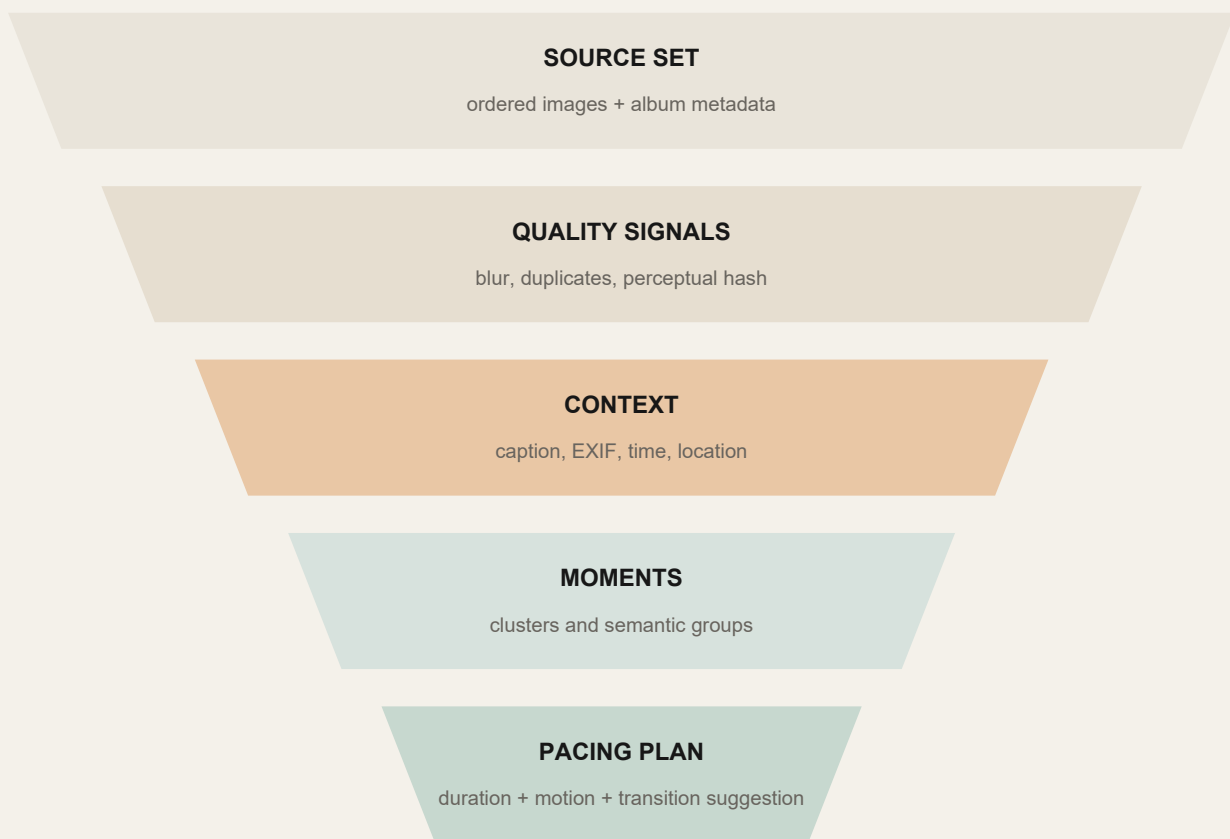
The pipeline is not one large model call. It is a chain of small contracts, persisted decisions, provider fallbacks, and a CPU-bound render job.



MEDIA INTELLIGENCE

The album is converted into evidence, not pixels

Vionto's media understanding is deliberately hybrid: inexpensive deterministic signals reduce noise, while vision-language captions supply semantic meaning for the story prompt.

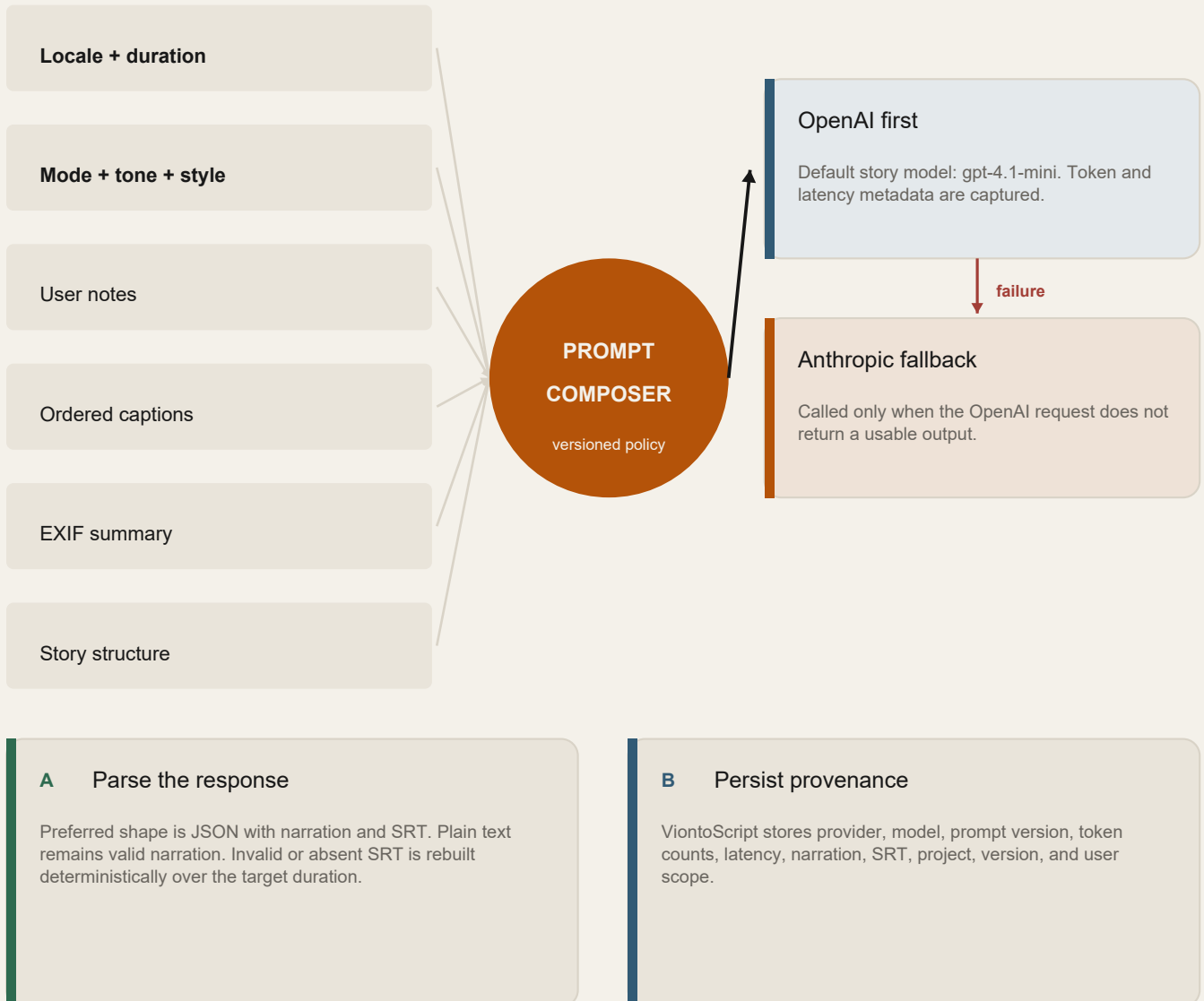


WHAT REACHES THE LLM

Ordered captions, album-item metadata, an EXIF summary, user notes, locale, mode, story mode, emotional tone, visual style, target duration, and optional story structure.

STORY SYSTEM

Prompt composition, provider fallback, and durable script output



CURRENT SCENE-PLAN REALITY

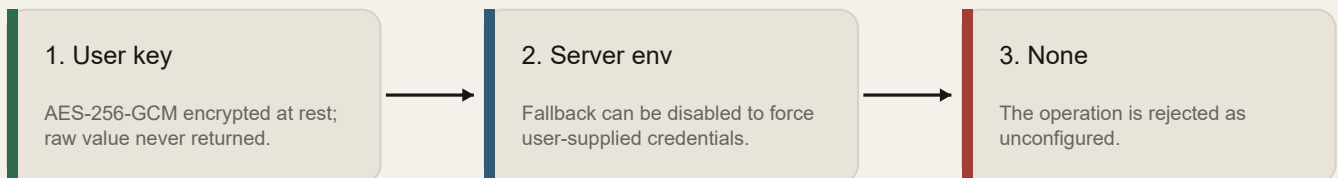
The provider interface defines a richer StoryPlan, but the current delegating adapter wraps generated narration in a single scene. Scene-level planning, local regeneration, and fine-grained caching are architectural targets, not fully realized production behavior.

PROVIDER LAYER

Capabilities are registered once and consumed through adapters

The registry is Vionto's catalogue of capability, auth shape, BYOK eligibility, implementation status, model tier, and rough cost. Factories return interfaces instead of exposing vendor calls.

PROVIDER	ANALYSIS	STORY	VOICE	VIDEO	STATUS
OpenAI	YES	YES	YES	-	LIVE
Anthropic	YES	YES	-	-	LIVE
ElevenLabs	-	-	YES	-	LIVE
Kling	-	-	-	YES	LIVE
fal.ai	-	-	-	YES	LIVE
Gemini	PLAN	PLAN	-	-	REGISTERED
Google TTS	-	-	PLAN	-	REGISTERED
FFmpeg	-	-	-	render	LIVE
Remotion	-	-	-	render	REGISTERED

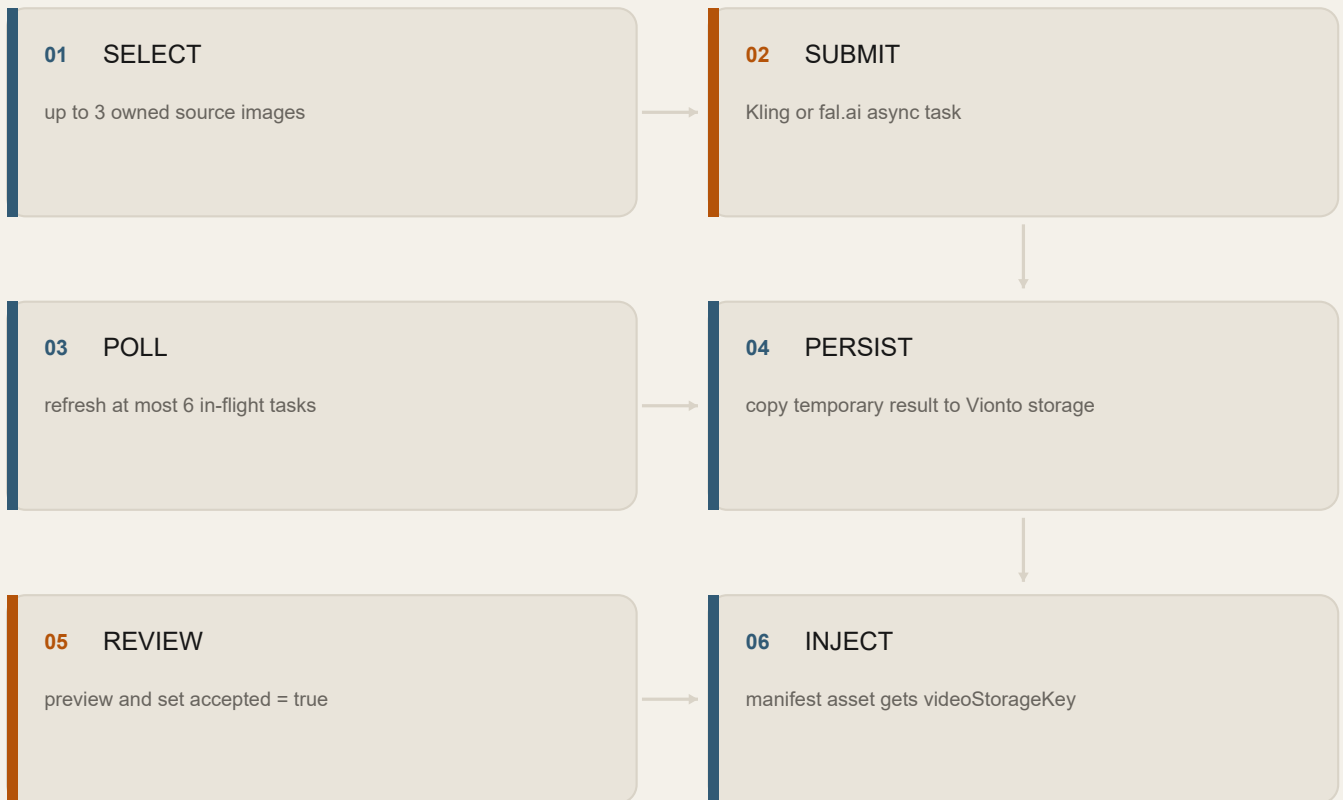
Credential resolution


Provider addition pattern: registry entry -> adapter -> factory case. The rest of the pipeline stays bound to AlbumAnalyzer, StoryPlanner, NarrationProvider, GenerativeVideoProvider, or VideoRenderer.

GENERATED MOTION

Premium AI video is an optional, review-gated substitution

Vionto does not generate an entire movie. At most three selected album images can be submitted per request for image-to-video generation. Each resulting clip lives beside the source asset and is inert until the user accepts it.



IDENTITY GUARDRAIL

A shared negative prompt suppresses distorted faces, identity changes, extra people or limbs, warped geometry, flicker, sudden movement, text, watermarks, and blur.

DETERMINISTIC FALLBACK

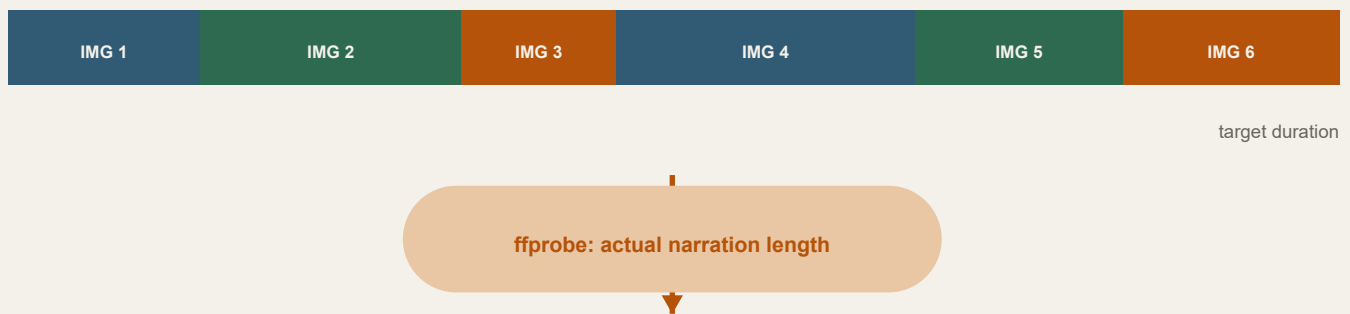
If no accepted clip exists, or a provider fails, the original image still renders with deterministic zoom/pan. The core export path does not depend on generative video availability.

NARRATION AND TIMING

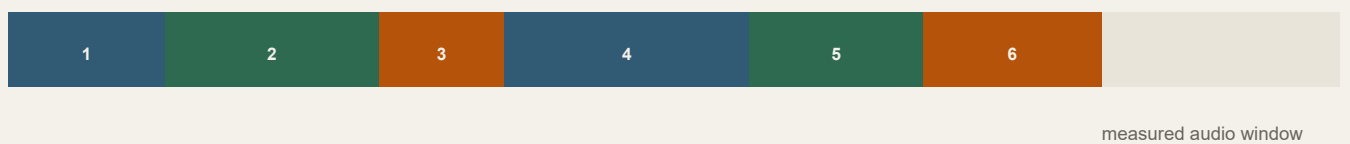
The soundtrack becomes the master clock

The requested duration is only a planning estimate. Once narration is synthesized or downloaded, fprobe measures its real duration. Vionto then rescales every image segment and subtitle cue into that measured window.

BEFORE TTS



AFTER TTS



Voice routing

Selected voice provider first; Azure, ElevenLabs, and OpenAI implementations exist.

Subtitle order

Stored SRT key, then inline SRT, then deterministic cues from narration.

Audio mix

Narration 1.0 + music 0.06; mix ends with narration in the active path.

EDGE CONDITION

Each image is clamped to at least one second. Very short narration combined with many images can therefore exceed the narration length slightly.

RENDER CONTRACT

The manifest is the seam between creative intent and execution

The web route rejects client-supplied manifests. After ownership and scope checks, it resolves authoritative project/version data, then validates the complete payload with Zod before placing it on the queue.

RENDER MANIFEST / SERVER BUILT

IDENTITY	<code>projectId versionId? userId jobId</code>
CREATIVE	<code>mode visualStyle storyMode emotionalTone</code>
OUTPUT	<code>duration aspect resolution fps codecs bitrates</code>
ASSETS [1..200]	<code>storageKey videoStorageKey? size duration motion transition</code>
AUDIO [0..16]	<code>narration music sfx source voice volume fades offset</code>
SUBTITLES	<code>text/key burn-in style timing SRT/VTT export</code>
STORY	<code>opening intro chapters climax closing dedication</code>
OVERLAYS	<code>time range asset index text kind placement style</code>
WORKER	<code>maxRetries [0..5] timeout [30..3600 sec]</code>

ZOD VALIDATED
OWNERSHIP BOUND
VERSION AWARE
QUEUE SAFE

RENDER WORKER

One job, six production stages, deterministic output

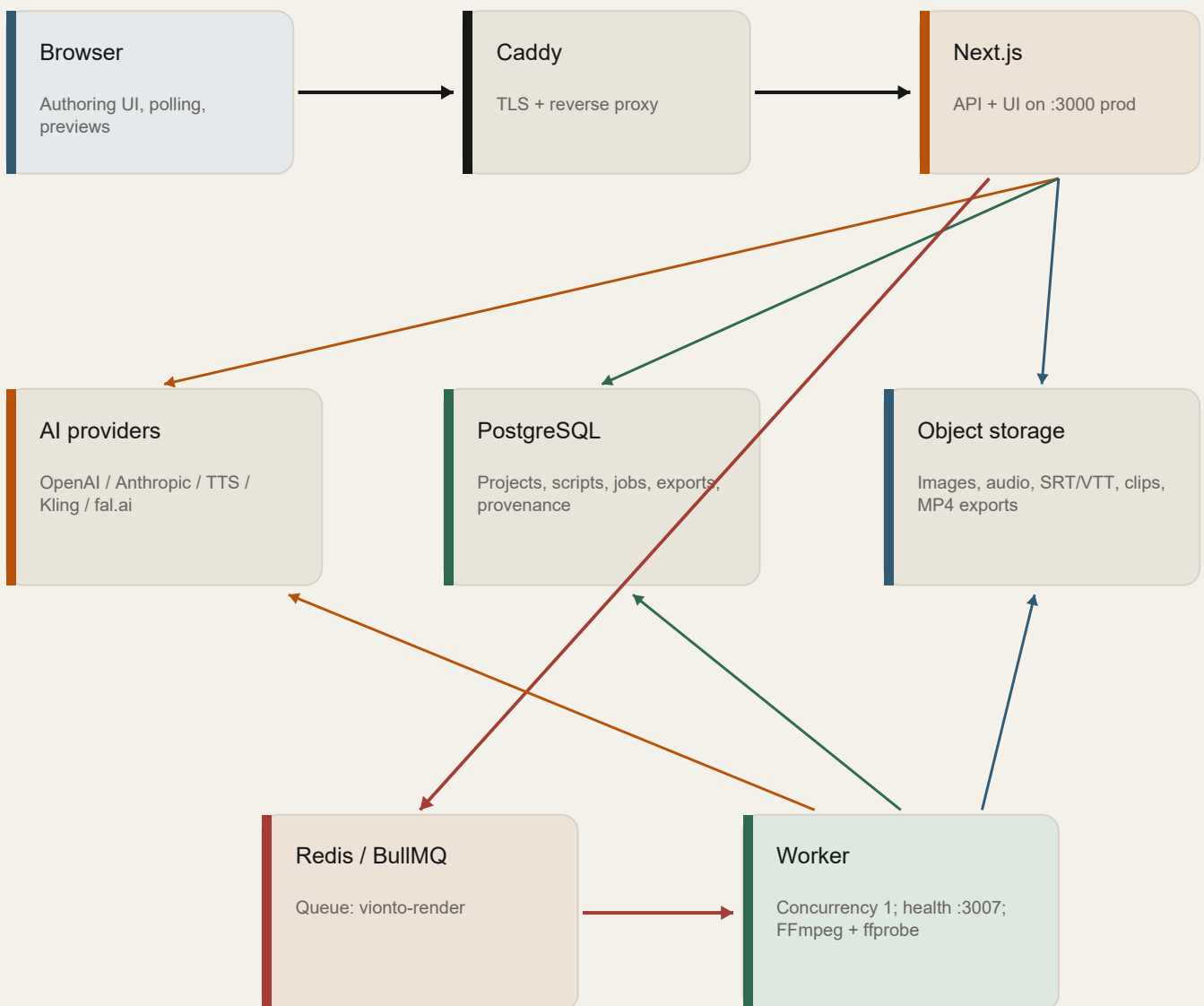
A dedicated BullMQ worker processes one CPU-heavy render at a time. It validates again, materializes remote objects in a per-job temporary directory, and records progress in PostgreSQL.



RUNTIME TOPOLOGY

Web requests and render work live in separate failure domains

Production is a small distributed system: Caddy fronts a standalone Next.js service; PostgreSQL persists domain and job state; Redis carries render work; object storage carries large media; the worker owns FFmpeg and TTS execution.

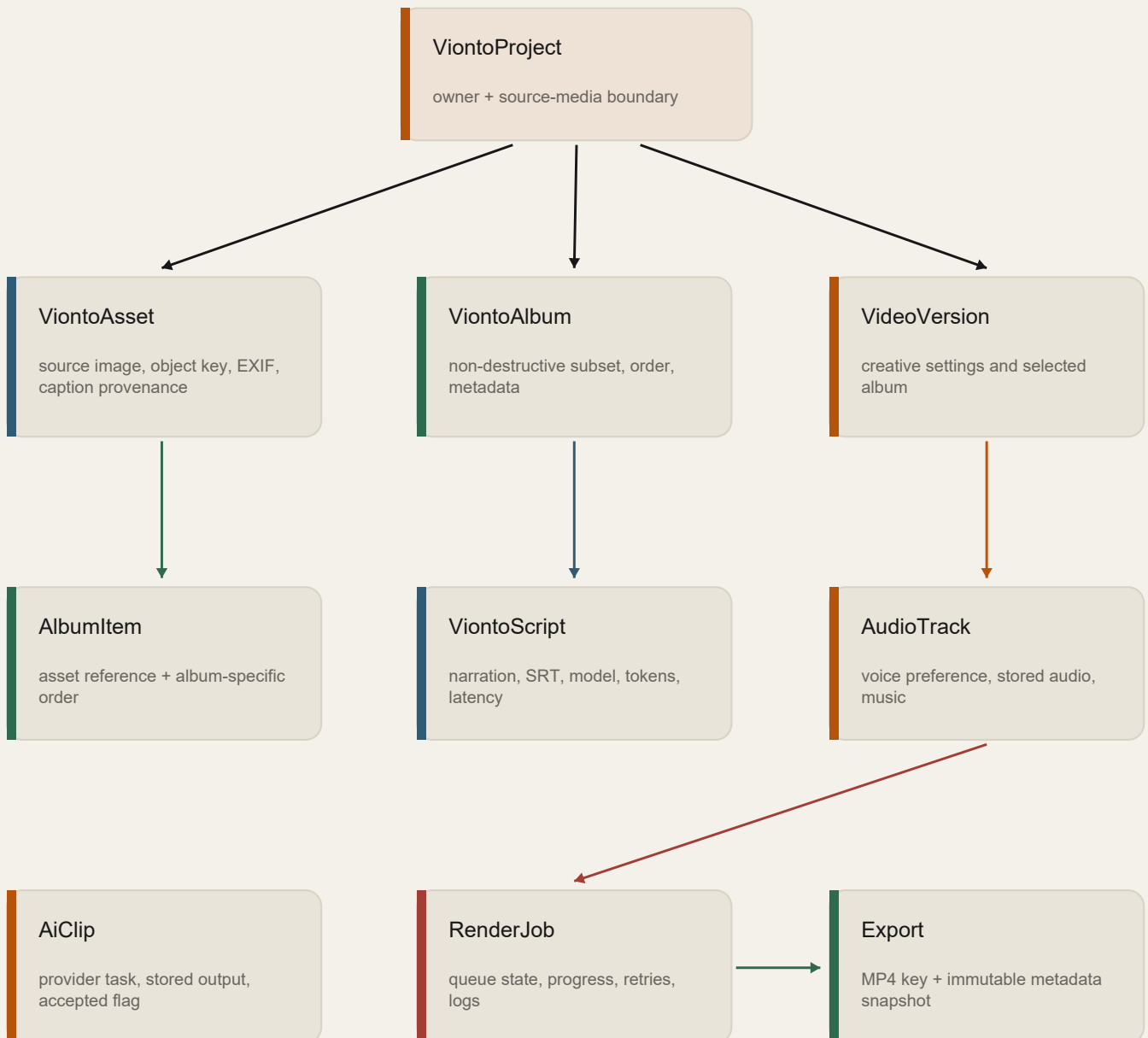


Local app start uses port 3004. The production standalone server is reverse-proxied as vionto:3000. The worker exposes a separate health endpoint on 3007.

DOMAIN MODEL

Project, album, and video version define three different boundaries

The data model separates ownership, curation, and creative configuration. This lets the same source photo appear in many albums and many video versions without duplicating its object-store payload.



AUTHORITATIVE CREATIVE SCOPE: when a versionId is supplied, version settings override project defaults. Effective media selection is explicit render album -> version album -> all project assets.

STATE AND PROGRESS

Three state vocabularies coexist - do not read the wrong one

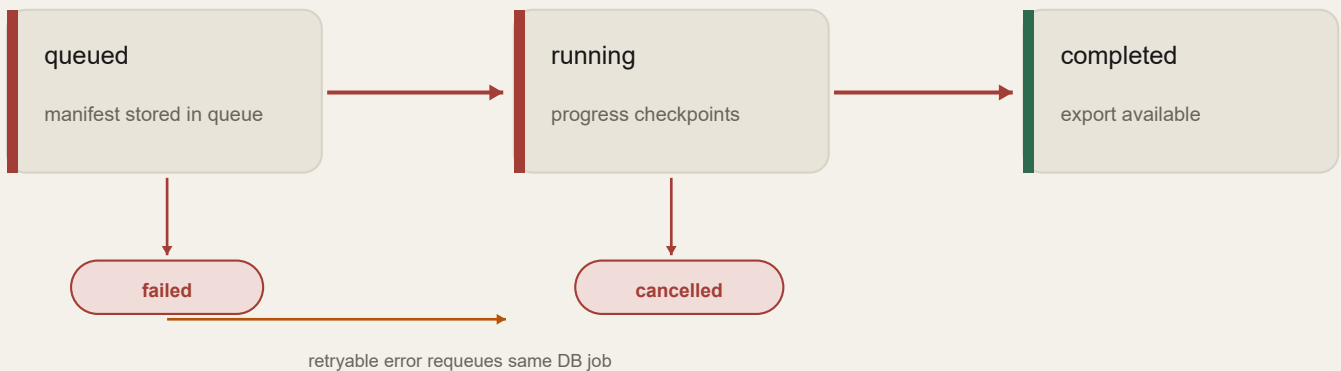
Project workflow, album lifecycle, and render jobs answer different questions. The canonical 15-state machine coexists with legacy project strings; live render truth remains on ViontoRenderJob.

CANONICAL PROJECT WORKFLOW

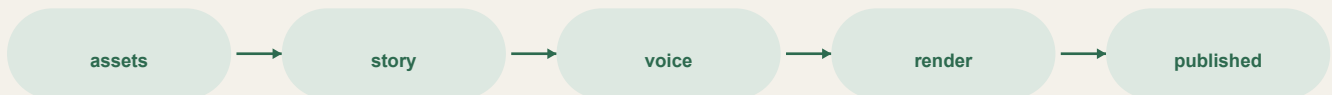


Any non-terminal live state may fail; edit paths can loop back from preview or completion.

RENDER JOB - OPERATIONAL TRUTH



ALBUM LIFECYCLE - USER MILESTONES



OPERATOR RULE

For live render status, use the render job. For user-facing workflow restoration, normalize the project state. For album completion milestones, use the monotonic album lifecycle.

RELIABILITY AND SECURITY

Controls exist at every expensive or trust-sensitive boundary

Controls concentrate at trust and cost boundaries: ownership, server-built manifests, schema validation, encrypted credentials, bounded polling, durable output copies, retries, cancellation checkpoints, and expiring downloads.

AUTH

JWT session + record ownership

Project and referenced assets are queried inside the authenticated user's scope.

INPUT

Zod and route constraints

Manifest, media counts, codecs, subtitle options, provider inputs, and durations are bounded.

SECRETS

AES-256-GCM BYOK

User keys win, environment keys may fall back, plaintext never returns to the UI.

QUEUE

Durable job state

Redis carries work; PostgreSQL carries state, retry count, progress, logs, and errors.

MEDIA

Owned namespace + short URLs

Source keys are ownership-bound; download and preview URLs expire; generated clips are copied.

FAILURE

Classified retry policy

TTS, timeout, network, and unknown errors retry; max retries defaults to three.

OBSERVABILITY TODAY

Provider / model / token / latency data is persisted for stories; job progress and logs are stored for renders; polling and SSE status endpoints exist. ViontoUsageMetric, plan quotas, and audit records are present in the schema, but the main media path does not yet meter or enforce them end to end.

IMPLEMENTATION REALITY

The gaps that matter most - ranked by production impact

A complete architecture document should show the unfinished seams. These are observable in the current repository and should shape the next hardening work.

PRIORITY	GAP	WHY IT MATTERS	RECOMMENDED MOVE
P0	Manual retry loses manifest	The retry route builds an invalid empty-asset payload.	Persist or reconstruct the original validated manifest.
P0	Upload sessions are in-memory	Restarts and multi-replica routing can lose session state.	Move sessions to Redis with expiry and ownership binding.
P1	Transitions are descriptive	Manifest supports them, but active concat rendering does not.	Implement xfade/slide filter graphs and duration math.
P1	Audio controls disconnected	Per-track volumes and advanced ducking are not used by final mix.	Wire active mix to stored settings; normalize loudness.
P1	Format contract mismatch	MOV/WebM are accepted but output is always MP4-specific.	Reject unsupported formats or finish codec/container routing.
P1	Storage cleanup gaps	Delete paths can leave source/export objects orphaned.	Add lifecycle jobs, reconciliation, and retention policy.
P2	Partial shared access	Editor permission does not traverse the full creation graph.	Centralize authorization across child resources.
P2	Quota/audit scaffolding only	Expensive calls are not comprehensively reserved or metered.	Add idempotent usage ledger and budget enforcement.

REFERENCE

Source map, operating notes, and glossary

This guide was derived from the current implementation and repository architecture notes. The files below are the fastest path for engineers who need to verify or extend a claim.

ARCHITECTURE	docs/vionto-architecture.md
AI DESIGN	apps/vionto/docs/ai-architecture.md
STORY ROUTE	apps/vionto/app/api/story/generate/route.ts
VISION + PROMPT	apps/vionto/lib/server/vision.ts; story-generation.ts
PROVIDER LAYER	apps/vionto/lib/server/ai/{registry, interfaces, adapters, factory, credentials}.ts
AI MOTION	apps/vionto/app/api/projects/[projectId]/ai-clips/**
PACING	apps/vionto/lib/server/{smart-pacing, pacing, image-scoring, image-clustering}.ts
MANIFEST	apps/vionto/lib/server/render-manifest.ts
RENDER API	apps/vionto/app/api/render/route.ts
WORKER	apps/vionto/worker.ts; lib/server/{queue, ffmpeg, tts, srt, audio-mix}.ts
STORAGE	apps/vionto/lib/server/storage.ts
DATA MODEL	packages/db/prisma/schema.prisma

GLOSSARY

Album. A reusable, ordered reference set over project assets.

Video version. A creative configuration and history boundary, usually linked to an album.

Manifest. The validated, server-built render contract consumed by the worker.

Accepted clip. A stored AI-generated video explicitly allowed to replace one still segment.

Deterministic render. The same validated inputs drive conventional FFmpeg composition and encoding.